

Back to Entropy: Evaluating Base Language Models Beyond Perplexity and Benchmarks

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Abstract

Choosing a base model today means reading benchmark scores that vendors optimise for, that can leak into training data, and that score a model before the fine-tune it will receive. We show that a few minutes of unlabelled adaptation to the domain predict how the models rank once fine-tuned for a task, before the task’s data exist. Every candidate is adapted the same way to fresh text from the target domain (minutes to an hour on one GPU) and ranked by how well it then predicts held-out text, in bits per byte (BPB). Ranked before any task exists, fifteen base models land within one rank of where they finish once fine-tuned on a news-writing task (Spearman rank correlation 0.99); HellaSwag follows at 0.97, while GSM8K and MMLU-Pro manage 0.59 and 0.71. Across seventeen models, eleven of them on four corpora (news, Reddit, Hacker News and arXiv mathematics), adaptation reorders the models, lifting some that look weak before adaptation to the top; the adapted ordering converges on HellaSwag, the likelihood-scored benchmark, turns toward GSM8K and MMLU-Pro when the text is mathematics, and stays stable across seeds, hardware and general-text corpora. The measurement needs no labels, no benchmark and no task. Code and artifacts are released.

1 Introduction

Consider an engineer choosing a base model. The product has a domain and plenty of its text; the shortlist is a dozen open checkpoints; the one chosen will be fine-tuned and shipped. The direct answer (fine-tune every candidate and compare the systems) costs a task definition, labels and a training run per model, so the choice is made from a scorecard instead: MMLU-Pro (knowledge questions), HellaSwag (picking the most natural ending of a passage, scored by likelihood) and GSM8K (grade-school maths), read as the answer to *how good is this model, for what we do?* This paper shows that the direct answer can be had before the task exists, from unlabelled text alone. Every candidate is adapted the same way, without labels, to text from the domain, and ranked by how well it then predicts held-out text from it, in bits per byte (BPB). That ranking matches the fine-tune’s almost exactly; of the scorecard, only HellaSwag comes close.

The scorecard misleads for two reasons. The first is that benchmarks are public. Each was written by other people, for other purposes, and released before the model existed; vendors optimise for the ones they report, and their items and subject matter can reach pre-training data. A high score is therefore consistent with a genuinely stronger model or with one that saw more of that exam, and because pre-training mixtures are not disclosed, no reader can tell which. Inside our own cohort, Llama-3.2-1B and Qwen-2.5-1.5B are 2.1 points apart on HellaSwag and 3.4% apart in adapted bits per byte, yet 55.3 points apart on GSM8K: two models that model fresh English almost identically well differ by nearly a factor of ten on one exam. The second reason is timing. Teams rarely deploy a base model exactly as it ships; they adapt it to their data first, so what matters is how good a model *becomes*. A model that has simply never seen prose like the target domain’s can look weak out of the box and still overtake one that looked stronger. Gemma-4-12B is the case in point: of the fifteen fine-tuned models it is ninth on GSM8K, eighth on MMLU-Pro and thirteenth

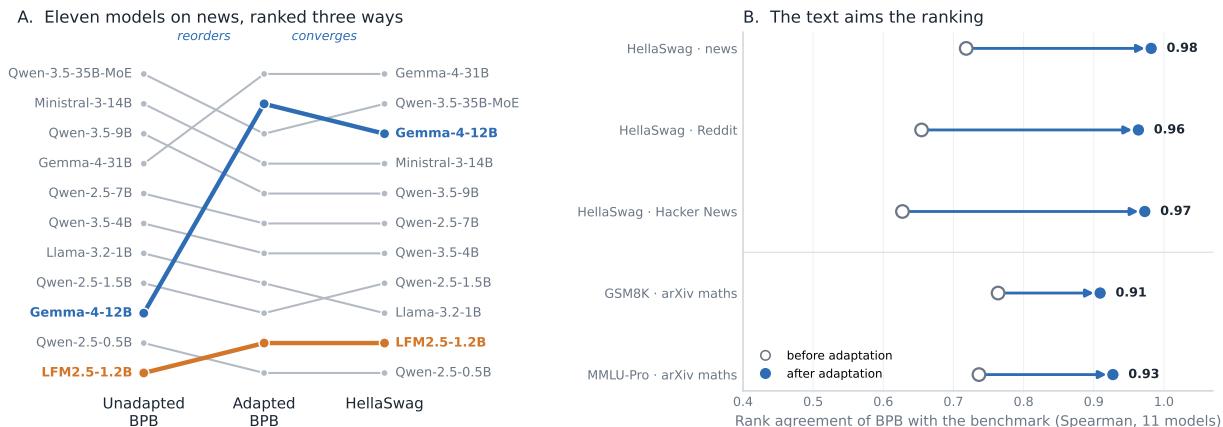


Figure 1: Adaptation reorders the models, and the text aims the new order. **A:** the eleven four-corpus models ranked by unadapted news BPB, by adapted news BPB and by HellaSwag. Between the first two columns the lines cross: adaptation reorders the models, lifting Gemma-4-12B from ninth to second and LFM2.5-1.2B out of last place. Between the last two they run almost parallel: the adapted order is HellaSwag’s. **B:** rank agreement (Spearman ρ , signed so that 1 means lower BPB goes with a higher score) between BPB and each benchmark, before adaptation (hollow) and after (filled). Adapted on general text, the ranking converges on HellaSwag on all three corpora; adapted on arXiv mathematics, it turns to GSM8K and MMLU-Pro. The full matrix is Table 4 in Appendix B.

on unadapted loss, second once adapted, and it builds the best of the fifteen fine-tuned systems (Figure 1A shows the same climb among eleven models).

Loss is the principled answer. There is no universal ground truth for which base model is “better”; a task tells us only which model is better at that task, and tasks disagree. Everything a base model can do, it learned by minimising cross-entropy on text, and adaptation minimises it again on the domain, so held-out loss is the objective every candidate was last trained on, measured out of sample. Cross-entropy is also a proper scoring rule: in expectation over text drawn from the domain’s distribution p , a model q scores $H(p, q) = H(p) + \text{KL}(p \| q)$, and $H(p)$ is the same for every model, so the difference between two models’ expected losses is *exactly* the difference in how far each is from the distribution that produced the text; held-out loss estimates it [1]. Taking that seriously asks three things of the measurement. The loss is counted in *bytes*, because every model counts in its own tokens and bytes are the one unit they all share. It is computed on *fresh, decontaminated* text, so that no candidate has trained on it. And it follows *equal adaptation*: every candidate is adapted to the domain by the same procedure, so each model is measured as it will be used rather than as it ships. Train-before-test [16] also prepares candidates identically before comparing them, but with labelled benchmark data; we do it unsupervised, on fresh text, and in bytes.

We apply the routine to seventeen base models from eight families (0.5B–35B): eleven on all four corpora, all seventeen on news, and the fifteen that fit one GPU fine-tuned on a news task (writing an article’s opening paragraph from its body, scored by ROUGE-L overlap with the original). The headline answers the engineer’s question. **Adapting to the domain predicts fine-tuning on the task.** Ranked before any task exists, adapted BPB orders fifteen models at Spearman 0.99 (rank correlation; 1 means identical orders) with where their fine-tuned systems finish: every model within one rank, eleven exactly. HellaSwag follows at 0.97; the reasoning benchmarks manage 0.59 (GSM8K) and 0.71 (MMLU-Pro) (Figure 4, Section 3.4). It is not news predicting news: adapted on Reddit instead, BPB orders the nine models that both analyses cover exactly as well (Spearman 0.983), while the same text unadapted manages about 0.6. Three findings explain why it works. **Adaptation reorders the models.** The unadapted (zero-shot) and adapted rankings agree only at Spearman 0.61–0.68 on general text, because unadapted loss mixes familiarity with a style of prose into capability; the only models that climb are the three adaptation helps most

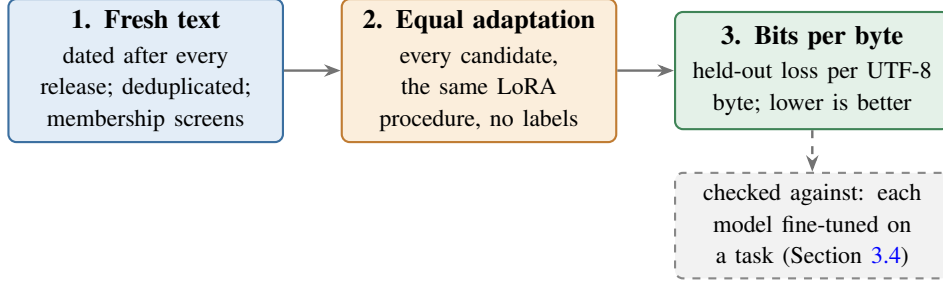


Figure 2: The protocol. The corpus is fixed and screened first, then every candidate is adapted the same way and scored on held-out text. The fine-tune of Section 3.4 is not part of the measurement; it is what the measurement is checked against.

(Figure 1A, Section 3.1). **The adapted ordering converges on the language benchmark, and can be aimed.** It matches HellaSwag at Spearman 0.96–0.98 on news, Reddit and Hacker News, up from 0.63–0.72, and adapted on mathematics it swings to GSM8K and MMLU-Pro (0.91 and 0.93; Figure 1B, Section 3.2). **It is stable where benchmarks are not.** Seed-to-seed variation is a thousandth of the spread between models, runs reproduce across hardware, and the ordering carries across general-text corpora, while identical weights move GSM8K by up to 0.69 points when only the host changes (Section 3.3). Section 2 gives the protocol and Section 3 the results. Every number recomputes from a committed artifact: <https://github.com/DavidAfonsoValente/entropy-bench>.

2 The Protocol

The measurement runs in three phases, and the order is load-bearing: the corpus is fixed and screened before any model touches it (Figure 2).

Phase 1: a decontaminated corpus. Evaluation text must be text no candidate has seen, and we guarantee it three ways. *Time*: the text is dated after every candidate’s release: Reddit and Hacker News are scraped in August 2026, after every model in the cohort shipped, and the arXiv titles and abstracts run from 9 June to 20 August. *Hygiene*: exact duplicates, text failing the Gopher quality heuristics [12] and MinHash/LSH near-duplicates at Jaccard 0.60 are removed, and the 80/10/10 split shares no identifier or normalised text across splits, so nothing in the test split reappears in training. *Membership screens*: the news snapshot (Google News, 8 June 2026, 119,054 articles) predates part of the release window, so every model is also tested for having seen it. Three reference-free tests look for the signatures memorised text leaves: Min-K++ [10] flags passages whose least likely tokens are unusually likely for the model; context sensitivity flags passages whose loss falls more than expected when the model is shown other text from the same corpus; and a perturbation test checks whether the model prefers the original passage over copies with swapped words. The *union* of everything any model flags, 80 passages, is removed for every model, so every candidate is measured on one identical corpus (Appendix C).

Phase 2: equal adaptation. Every candidate is adapted to the retained training text, packed into contiguous 512-token blocks, with LoRA [28] on all linear layers, and by the same procedure. In the full method each candidate’s hyperparameters come from its own Optuna search [25]: a Tree-structured Parzen Estimator proposes the learning rate, batch size, warm-up, weight decay, LoRA rank, α and dropout; successive halving prunes weak trials from 50 steps on; and every trial is scored by BPB on the validation split, never the test split. Every model receives the same number of trials and no time limit, because a search capped by wall-clock time lets small models finish more trials than large ones and so tunes them better, a size bias

we observed directly (Appendix C). The chosen configuration is then trained until validation BPB stops improving. The released pipeline implements this (`--hparams sweep --no-time-limit`).

For compute, the results reported here use one hand-picked configuration for every candidate instead (LoRA rank 16, $\alpha=32$, dropout 0.05, learning rate 10^{-4} , effective batch 32) and a fixed 250 update steps. That is 4.1M training tokens per candidate, which on one A100 cost 3.7 GPU-minutes at 0.5B, 11.6 at 7.7B, 24.7 at 12B and 62.7 at 31B. The configuration sits where our own earlier search settled (rank 16 won for eight of the eleven models and the median winning learning rate was 1.1×10^{-4}), and its endpoint is stable to seed and hardware (Section 3.3).

Phase 3: bits per byte. Cross-entropy on the held-out test split is divided by the UTF-8 byte length $|B|$ of the text those predictions cover:

$$\text{BPB} = \frac{\mathcal{L}_{\text{nats}}}{\ln 2} \times \frac{|\text{Tokens}|}{|B|} \quad (1)$$

The byte count is a property of the text and identical for every candidate, so a larger vocabulary earns no advantage the way it would under per-token perplexity. Test blocks are independent and non-overlapping, so context resets at every cut, and the beginning-of-document tokens inserted between packed documents stay available as context but are never scored. We report both endpoints, before adaptation (unadapted, or zero-shot, BPB) and after, because the difference between them is itself informative (Section 3.1).

Cohort and setup. We evaluate 11 base checkpoints spanning dense Transformers, one sparse mixture-of-experts and one non-Transformer (LiquidAI’s LFM2.5-1.2B) (0.5B–35B parameters; Table 9) on all four corpora, and extend the news cohort to 17 with six models chosen in advance to fill gaps in size and add publishers (Table 8), from the Llama, Qwen, Gemma, Ministral, LiquidAI, Falcon, Mistral and SmolLM families [29, 30, 31, 32, 33]. For comparison every model is also scored on MMLU-Pro [19], HellaSwag [20] and GSM8K [21] with *lm-evaluation-harness* 0.4.12 [13] on one host and one build, under the protocol of Appendix F, directly observed scores rather than vendor-reported card values, with the harness logs published so each recomputes from raw output. Adaptation ran on Leonardo (CINECA), 3 nodes $\times$ 4 A100 64 GB. Most comparisons are scored by *pairwise accuracy*: the fraction of model pairs that a ranking orders the same way as a reference. Its 95% intervals come from a bootstrap that resamples models, never pairs, because pairs that share a model are not independent (20,000 draws); intervals on one fine-tuned pair resample its 500 evaluation articles. Pairwise accuracy counts only *size-matched* pairs, two models whose parameter counts differ by less than a factor of two: when one model is more than twice the other’s size it usually wins, and a ranking would get that pair right just by following size.

3 Results

3.1 Adaptation reorders the models

Adapting every model before measuring does not just lower every score; it reorders the models (Figure 1A, Table 1). On news the unadapted and adapted rankings of the eleven models agree only at Spearman 0.655, on Reddit 0.609 and on Hacker News 0.682, and a model can move seven places. The movement is not spread evenly. Adaptation lowers loss by 3.6–5.4% on news for every Qwen, Llama and Ministral model, but by 17.0% for Gemma-4-31B, 24.5% for Gemma-4-12B and 36.4% for LFM2.5-1.2B, and those three are the only models that move up; every other model slips one or two places to make room. Gemma-4-12B is ninth of eleven unadapted and second adapted, a 12B model the unadapted reading puts below two models under 2B; LFM2.5-1.2B, the one non-Transformer, is dead last unadapted at 1.34 bits per byte against a 0.5B Qwen’s 0.93, and ahead of it after adaptation.

Model	Params (B)	News BPB (rank)		Moves	Gain	Benchmark score, % (rank)		
		unadapted	adapted			HellaSwag	GSM8K	MMLU-Pro
Gemma-4-31B	31.4	0.752 (4)	0.624 (1)	↑3	17.0%	87.9 (1)	85.0 (3)	57.1 (2)
Gemma-4-12B	12.0	0.866 (9)	0.653 (2)	↑7	24.5%	84.0 (3)	64.8 (7)	44.8 (7)
Qwen-3.5-35B-MoE*	34.7	0.707 (1)	0.678 (3)	↓2	4.1%	85.6 (2)	86.6 (1)	61.4 (1)
Ministral-3-14B	14.0	0.718 (2)	0.679 (4)	↓2	5.4%	82.9 (4)	80.8 (6)	52.8 (4)
Qwen-3.5-9B	9.0	0.740 (3)	0.708 (5)	↓2	4.4%	81.6 (5)	85.8 (2)	55.6 (3)
Qwen-2.5-7B	7.7	0.758 (5)	0.717 (6)	↓1	5.4%	80.3 (6)	81.3 (4)	47.2 (6)
Qwen-3.5-4B	4.2	0.782 (6)	0.749 (7)	↓1	4.2%	77.2 (7)	81.0 (5)	52.4 (5)
Llama-3.2-1B	1.2	0.811 (7)	0.771 (8)	↓1	4.9%	65.8 (9)	6.4 (11)	11.2 (11)
Qwen-2.5-1.5B	1.6	0.833 (8)	0.797 (9)	↓1	4.3%	68.0 (8)	61.6 (8)	25.6 (8)
LFM2.5-1.2B	1.2	1.340 (11)	0.852 (10)	↑1	36.4%	60.9 (10)	54.1 (9)	16.4 (9)
Qwen-2.5-0.5B	0.5	0.935 (10)	0.901 (11)	↓1	3.6%	51.8 (11)	34.3 (10)	16.0 (10)

Table 1: The eleven models, ranked every way. Ordered by adapted news BPB (lower is better). *Moves* is how many places adaptation shifts a model; *Gain* is the loss reduction it buys. Three models gain far more than the rest (the two Gemmas and LFM2.5-1.2B) and they are the only three that climb. Once adapted, the order is HellaSwag’s almost exactly, while GSM8K and MMLU-Pro scatter: Llama-3.2-1B, eighth by BPB and ninth on HellaSwag, is last on both. *Qwen-3.5-35B-MoE activates 3B of its 34.7B parameters.

The reason is familiarity. Unadapted loss conflates two things that need separating: *this model already writes like my corpus* and *this model is a capable model*. Figure 3 shows the confound in the unadapted loss, resolved by position inside a packed evaluation block. Because blocks are independent, every block is a cold start. Unadapted, Gemma-4-12B opens a block at 2.83 bits per byte where Qwen-2.5-1.5B (eight times smaller) opens at 2.04, and it stays behind for roughly the first half of the block; by the end of the block it has passed it, 0.72 against 0.77. A block-averaged number therefore reports mostly how quickly a model finds its footing in unfamiliar prose (the document conventions, formatting and register of this corpus), which is a real property but not the one a choice of model turns on. A short equal adaptation supplies exactly the missing familiarity, and the reduction it buys is largest in the opening but does not vanish after it: what is left is closer to capability. Unadapted loss is not wrong, then, but it answers a different question (how much this model already resembles the target text), and the models it penalises most are those whose pre-training prose differs most from it, which is also why a non-Transformer with its own training mixture is hit hardest.

3.2 The adapted ordering converges on the language benchmark, and can be aimed

Of the three public benchmarks, HellaSwag is the one closest to language modelling: it asks which ending of a passage is most natural, and the harness scores it by the likelihood of that ending. Before adaptation, BPB on news, Reddit and Hacker News ranks the eleven models at Spearman 0.63–0.72 with HellaSwag; after adaptation the same rank correlation is 0.96–0.98, and on news the adapted ranking orders 53 of the 55 model pairs the same way as HellaSwag (Table 1; the full matrix is Table 4 in Appendix B). Three independently collected corpora arrive at the same ordering once familiarity is removed, and it is the ordering of the benchmark that measures the same thing. Because HellaSwag and BPB are both likelihoods, this is consistency rather than a second opinion: measuring on fresh text of one’s own gives up nothing the likelihood benchmark captures, while escaping everything that makes a public benchmark suspect.

The effect is specific to what the text is about. On general text the correlation with GSM8K falls on all three corpora, and with MMLU-Pro on two of them; adapted on arXiv mathematics instead, BPB swings toward the reasoning benchmarks (Spearman 0.91 with GSM8K and 0.93 with MMLU-Pro, up from 0.76 and 0.74), and ranking by mathematics BPB picks GSM8K’s own top model. The choice of text points the measurement at a capability: the same instrument, fed news, recovers the language benchmark, and fed

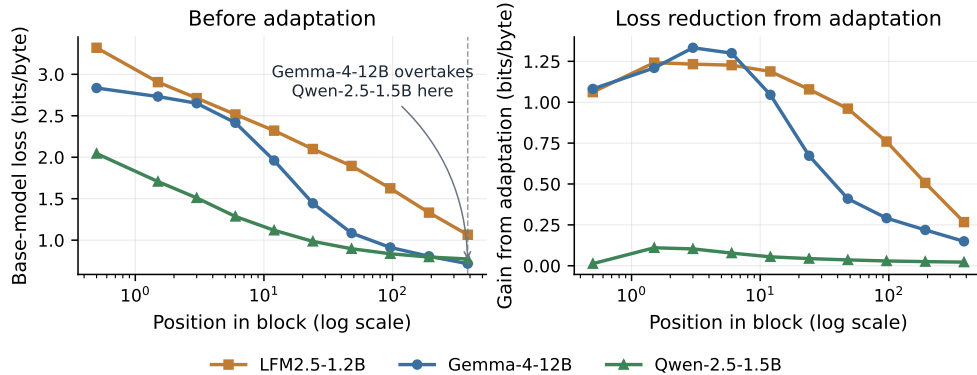


Figure 3: Why the unadapted number mis-ranks: it is largely scoring cold starts. *Left:* unadapted loss by position within a packed 512-token evaluation block, for Gemma-4-12B, Qwen-2.5-1.5B and LFM2.5-1.2B. Unadapted, Gemma-4-12B opens a block at 2.83 bits per byte against Qwen-2.5-1.5B’s 2.04 (a 12B model losing to a 1.5B one) and closes it at 0.72 against 0.77. A block average over-weights the opening, and that is *part* of what puts Gemma-4-12B ninth of eleven on unadapted BPB. *Right:* the loss reduction adaptation buys at each position is largest in the opening but does not vanish after it: adaptation teaches the corpus’s opening conventions, and does more besides. The right panel uses the three adapters of the earlier per-model search (Appendix C), so its gains differ from the fixed-recipe ones in Table 1; the pattern by position is what it shows.

mathematics, recovers the reasoning ones. Benchmarks cannot be aimed this way, because each is fixed; a corpus is chosen by whoever is choosing the model.

3.3 It is stable where benchmarks are not

A ranking is only as useful as it is repeatable. Three preregistered seed replicates of all 17 news runs put the median within-model standard deviation of adapted BPB at 0.000087 against a between-model spread of 0.074340, a ratio of 0.0012, and not one of the 40 size-matched pairs reverses its order across seeds; the one model where seed noise is visible at all is Gemma-4-12B, by up to 1.24%. Re-running two models on different hardware reproduced unadapted BPB exactly and adapted BPB to within 0.0003, and a run repeated through a different software stack reproduced to 0.01%. The ordering also carries across domains: adapted rankings agree at $\rho = 0.991$ between news and Reddit, 0.955 between news and Hacker News and 0.964 between Reddit and Hacker News (Table 6 and Figure 5, Appendix C). The unadapted ordering is just as stable across domains (0.982, 0.964 and 0.973), so adaptation does not buy stability; it buys a *different* stable ordering, the one Section 3.1 explains and Section 3.2 shows converging on the language benchmark. Only mathematics breaks from the rest, and only after adaptation: its agreement with the general corpora falls from 0.91–0.97 unadapted to 0.74–0.84 adapted, because adapting on mathematics exposes a different capability.

Benchmark scores do not have this property. Identical released weights move -0.60 to $+0.69$ GSM8K points when only the host changes, A100 to H100, and one decoding default a checkpoint ships with was worth 9.86 GSM8K and 14.30 MMLU-Pro points to Gemma-4-12B in our re-score (Appendix F). The difference is structural rather than a matter of care. Both measurements run on hardware with the same floating-point nondeterminism (GPU reductions are not associative, and their results can change with batch size and kernel choice [26]), but they are exposed to it in opposite ways. A generative benchmark decodes greedily: a perturbation in the last bits of two nearly tied logits flips one token, the model then conditions on its own altered output, and the divergence compounds over a chain-of-thought answer until it flips a right-or-wrong verdict, with each of GSM8K’s 1,319 questions worth 0.076 points. BPB is scored with teacher forcing: every prediction conditions on the true text rather than on the model’s own earlier choices,

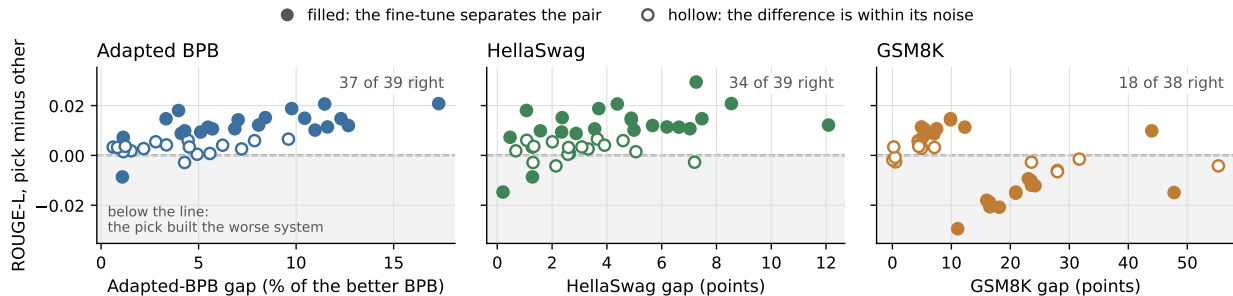


Figure 4: Does each way of ranking agree with the fine-tuned outcome? Each point is one pair of the fifteen fine-tuned models whose parameter counts are within $2\times$ of each other: 39 such pairs, 38 for GSM8K, on which two models tie. Each pair is oriented so that the ranking prefers its first model: x is how strongly it prefers it (the gap in that measure), y is how much higher the first model’s fine-tuned system scores in ROUGE-L. Points above the line are pairs the ranking orders as the fine-tune does; filled markers are *resolved* pairs, whose ROUGE-L difference has a 95% bootstrap interval over the 500 articles that excludes zero. Adapted BPB agrees on 37 of 39, and on 21 of the 22 resolved pairs; HellaSwag on 34; GSM8K on 18 of 38, and every one of its eleven gaps between 15 and 25 points sits below the line.

so a perturbation shifts one log-probability slightly and never propagates, and it is averaged over about 1.3 million token predictions on news. Decoding settings, which moved Gemma-4-12B by ten GSM8K points, do not enter at all, because nothing is generated. A generative benchmark measures the model, the harness, the decoding configuration and the hardware at once; bits per byte on a fixed corpus, after a fixed adaptation, measures the model.

3.4 Adapting to the domain predicts fine-tuning on the task

Whichever model a team chooses, it will then train that model for its task. The practical question is whether the cheap measurement already says how that will go. We fine-tuned fifteen of the seventeen models of the extended cohort (Table 8), all but Gemma-4-31B and Qwen-3.5-35B-MoE, which do not fit one GPU, on a news task, writing the opening paragraph of an article from its body, with one identical LoRA recipe and ROUGE-L against the human-written lede (the article’s opening paragraph) on 500 held-out articles from a split the BPB test blocks do not touch; the analysis was fixed in code before the jobs ran (Appendix D). The task is chosen so that every article supplies its own label: the journalist’s opening paragraph is the reference, so no annotation is needed, and writing it requires reading the whole article and producing prose in the domain’s style. ROUGE-L, the longest-common-subsequence overlap with that reference, is the standard automatic score for such summary generation [27]. Over all 39 size-matched pairs the fifteen systems form, adapted BPB, measured with no task and no labels, orders 0.949 of the pairs as the fine-tune does (95% interval [0.817, 1.000]); HellaSwag orders 0.872, while GSM8K and MMLU-Pro sit at 0.474 and 0.513, a coin (Figure 4, Table 2). On the resolved pairs, those the fine-tune itself separates, it agrees on 21 of the 22, and its ranking of the fifteen correlates with their final order at Spearman 0.99, every model within one rank and eleven exactly, where unadapted BPB manages 0.71, GSM8K 0.59, MMLU-Pro 0.71 and HellaSwag 0.97. It also beats ranking by size under this recipe, by +0.231 in pairwise accuracy [+0.059, +0.452]. A deliberately different fine-tuning recipe, recipe B (four times the rank, three times the learning rate, half the batch, a new seed) leaves the systems’ ordering almost unchanged (Spearman 0.982) and adapted BPB still first at 0.923, in agreement on all 20 pairs that recipe separates. The two measurements share a domain, so their agreement could be suspected of being news predicting news. It is not: adapted on Reddit instead, BPB orders the nine models that were both fine-tuned and adapted on all four corpora exactly as well as news BPB does (Spearman 0.983 each), Hacker News reaches 0.933, and the same corpora unadapted reach only about 0.6. What carries the prediction is adaptation to fresh general text, not the task’s own domain; arXiv

mathematics, which aims the ranking at reasoning instead (Section 3.2), predicts the news fine-tune less well. That is the point: the few minutes of unlabelled adaptation behind a BPB number already contain most of what fine-tuning on the task reveals, and the reasoning benchmarks read before choosing contain little of it. **Gemma-4-12B**, 9th and 8th of fifteen on GSM8K and MMLU-Pro, builds the best system of the fifteen, beating Ministral-3-14B by +0.018 ROUGE-L [+0.011, +0.025]; and the 55.3-point GSM8K gap between Llama-3.2-1B and Qwen-2.5-1.5B buys no measurable difference once both are fine-tuned. Every model improved on its own base weights and cleared what copying the article’s first sentence scores (ROUGE-L 0.125).

	All 39 pairs	95% CI	22 resolved pairs	Recipe B
Adapted BPB	0.949	[0.817, 1.000]	0.955	0.923
HellaSwag	0.872	[0.640, 1.000]	0.909	0.846
Unadapted BPB	0.769	[0.444, 1.000]	0.682	0.744
Parameter count	0.718	[0.463, 0.912]	0.773	0.744
MMLU-Pro	0.513	[0.250, 0.846]	0.500	0.538
GSM8K	0.474	[0.194, 0.824]	0.455	0.447

Table 2: Which selector predicts the fine-tuned system? Fifteen models, each fine-tuned identically on the lede task; the outcome is each system’s ROUGE-L on 500 held-out articles. Each row is a selector’s pairwise accuracy over every size-matched pair, with the model-level cluster bootstrap, and its accuracy on the 22 pairs whose fine-tuned difference is itself resolved by a paired bootstrap over articles. The last column scores the same selectors against systems built with the deliberately different recipe B. GSM8K ties one pair and is scored on 38.

4 Conclusion

Adapting to the domain predicts fine-tuning on the task: minutes of unlabelled adaptation tell which base model will fine-tune best. All fifteen fine-tuned systems finish within one rank of where adapted BPB placed their models (Spearman 0.99), where the reasoning benchmarks manage 0.59–0.71, and adapting on Reddit instead of news predicts the news fine-tune just as well. The principle explains why: there is no universal ground truth, the loss is the objective the adaptation minimises for every candidate, and its differences are differences in distance from the text. Adaptation removes familiarity with the text’s style, the ordering that remains follows the capability the text exercises (language on general text, reasoning on mathematics), and it is stable where benchmark scores move with the host.

Fine-tuning every candidate evaluates one task once its data and budget exist; adapted BPB gives the fine-tune’s ranking before either, in minutes per model, from text alone. The method tunes each model; for compute, the runs here share one recipe, and a fine-tune with four times the rank and three times the learning rate leaves the systems’ order at Spearman 0.982, with adapted BPB predicting it under both. HellaSwag comes close on news, but it is one fixed, public exam: on mathematics adapted BPB turns to the reasoning benchmarks, and the decontaminated text it is measured on is text no model has seen. Every checkpoint here is a base model; a post-trained one is adapted the same way, and the measurement runs on it unchanged.

Adaptation here is 4.1M tokens per model, minutes to an hour on one A100. A different seed moves adapted BPB by up to 1.24% (Gemma-4-12B; the median model, 0.01%), so we treat a gap under 2% as a tie. The unadapted number costs one forward pass and mostly measures how familiar a model already is with the text. Results are comparable only when corpus, split, context, masking and adaptation all match, so the released benchmark keeps one leaderboard per corpus and recipe: a news track run by the maintainers, and a mathematics track on arXiv titles and abstracts whose corpus is public.

References

- [1] Shannon, C. E. (1951). Prediction and entropy of printed English. *Bell System Technical Journal*, 30:50–64.
- [2] Mielke, S. J. (2019). Can you compare perplexity across different segmentations? Technical note.
- [3] Gao, L., et al. (2021). The Pile: An 800GB Dataset of Diverse Text for Language Modeling. *arXiv:2101.00027*.
- [4] Ahia, O., et al. (2023). Do All Languages Cost the Same? Tokenization in the Era of Commercial Language Models. *EMNLP*. *arXiv:2305.13707*.
- [5] Magnusson, I., et al. (2023). Paloma: A Benchmark for Evaluating Language Model Fit. *arXiv:2312.10523*.
- [6] Carlini, N., et al. (2021). Extracting Training Data from Large Language Models. *USENIX Security*.
- [7] Carlini, N., et al. (2022). Quantifying Memorization Across Neural Language Models. *ICLR 2023*. *arXiv:2202.07646*.
- [8] Lee, K., et al. (2022). Deduplicating Training Data Makes Language Models Better. *ACL*. *arXiv:2107.06499*.
- [9] Shi, W., et al. (2023). Detecting Pretraining Data from Large Language Models. *ICLR 2024*. *arXiv:2310.16789*.
- [10] Zhang, J., et al. (2024). Min-K%++: Improved Baseline for Detecting Pre-Training Data from Large Language Models. *ICLR 2025*. *arXiv:2404.02936*.
- [11] Xu, C., et al. (2024). Benchmark Data Contamination of Large Language Models: A Survey. *arXiv:2406.04244*.
- [12] Rae, J. W., et al. (2021). Scaling Language Models: Methods, Analysis & Insights from Training Gopher. *arXiv:2112.11446*.
- [13] Gao, L., Biderman, S., et al. (2024). A Framework for Few-Shot Language Model Evaluation (lm-evaluation-harness). EleutherAI.
- [14] Heineman, D., Hofmann, V., Magnusson, I., et al. (2025). Signal and Noise: A Framework for Reducing Uncertainty in Language Model Evaluation. *NeurIPS 2025*.
- [15] Thrush, T., Potts, C., and Hashimoto, T. (2024). Improving Pretraining Data Using Perplexity Correlations. *arXiv:2409.05816*.
- [16] Zhang, G., Dominguez-Olmedo, R., and Hardt, M. (2025). Train-before-Test Harmonizes Language Model Rankings. *arXiv:2507.05195*.
- [17] Ruan, Y., Maddison, C. J., and Hashimoto, T. (2024). Observational Scaling Laws and the Predictability of Language Model Performance. *NeurIPS 2024*.
- [18] Gadre, S. Y., Smyrnis, G., Shankar, V., et al. (2025). Language Models Scale Reliably with Over-Training and on Downstream Tasks. *ICLR 2025*.
- [19] Wang, Y., et al. (2024). MMLU-Pro: A More Robust and Challenging Multi-Task Language Understanding Benchmark. *NeurIPS D&B*. *arXiv:2406.01574*.
- [20] Zellers, R., et al. (2019). HellaSwag: Can a Machine Really Finish Your Sentence? *ACL*. *arXiv:1905.07830*.
- [21] Cobbe, K., et al. (2021). Training Verifiers to Solve Math Word Problems (GSM8K). *arXiv:2110.14168*.
- [22] Krause, B., Kahembwe, E., Murray, I., and Renals, S. (2018). Dynamic Evaluation of Neural Sequence Models. *ICML 2018*. *arXiv:1709.07432*.
- [23] Krause, B., Kahembwe, E., Murray, I., and Renals, S. (2019). Dynamic Evaluation of Transformer Language Models. *arXiv:1904.08378*.
- [24] Gururangan, S., et al. (2020). Don’t Stop Pretraining: Adapt Language Models to Domains and Tasks. *ACL*. *arXiv:2004.10964*.
- [25] Akiba, T., Sano, S., Yanase, T., Ohta, T., and Koyama, M. (2019). Optuna: A Next-generation Hyperparameter Optimization Framework. *KDD*. *arXiv:1907.10902*.
- [26] He, H., and Thinking Machines Lab (2025). Defeating Nondeterminism in LLM Inference. Thinking Machines Lab: Connectionism. <https://thinkingmachines.ai/blog/defeating-nondeterminism-in-llm-inference/>.
- [27] Lin, C.-Y. (2004). ROUGE: A Package for Automatic Evaluation of Summaries. *ACL Workshop on Text Summarization* Branches Out.
- [28] Hu, E. J., et al. (2021). LoRA: Low-Rank Adaptation of Large Language Models. *ICLR 2022*. *arXiv:2106.09685*.
- [29] Dubey, A., et al. (2024). The Llama 3 Herd of Models. *arXiv:2407.21783*.
- [30] Yang, A., et al. (2024). Qwen2.5 Technical Report. *arXiv:2412.15115*.
- [31] Gemma Team (2026). Gemma 4 Technical Report. *arXiv:2607.02770*.
- [32] Liu, A. H., et al. (2026). Ministral 3. *arXiv:2601.08584*.
- [33] Qwen Team (2026). Qwen3.5: Towards Native Multimodal Agents. Official release and model card. <https://qwen.ai/blog?id=qwen3.5>.

A Related Work

From perplexity to comparable loss. Language modelling was grounded in byte- and character-level entropy from the start [1], but token-level perplexity depends on each model’s vocabulary [2, 4]; normalising by UTF-8 bytes restores a common unit [3, 5], and loss-based metrics carry better signal and lower noise than accuracy across 465 models [14]. Our contribution is the operational protocol around the metric, not the metric.

Where the loss is measured matters. Thrush et al. turn the varying correlation between per-domain perplexity and benchmark performance into a pretraining-data selection rule [15]. Their estimand is the complement of ours: they fix the benchmark suite and choose data to *train* one strong model, where we fix the candidate pool and ask how the *selection ranking* moves.

Equal preparation. Dynamic evaluation updates a model on recent test history [22, 23] and domain-adaptive pretraining improves fit to a target distribution [24]. Closest to our motivation, *train-before-test* gives 61 models identical supervised preparation across 24 tasks and raises cross-benchmark Kendall agreement from 0.52 to 0.76 [16]; they rank post-fine-tuning accuracy on public labelled benchmarks, where we rank held-out fit after *unsupervised* adaptation to a fresh corpus, across tokenizers. The core idea, prepare every candidate identically before comparing them, is theirs; what we add is doing it without labels, on text no vendor has seen, in a unit that compares tokenizers, and checking the result against real fine-tuned systems. Controlled families show a smooth loss-to-error relation [18, 17]; we test it *across* families.

Contamination. Models memorise duplicated pretraining text [6, 7], so deduplication and contamination analysis are necessary [8, 11]; reference-free membership signals are useful screens [9, 10] but prove nothing about a proprietary training set, so we use them only for conservative quarantine.

Corpus	Raw records	Documents	Median words	Text	Contamination control
Google News, 8 Jun 2026	119,054	119,054	362	361 MB	Min-K++ / context / perturbation union
Reddit, 6–10 Aug 2026 [†]	3,054,623	141,527	624	533 MB	Post-release scrape
Hacker News, 6–10 Aug 2026	60,814	6,086	690	46 MB	Post-release scrape
arXiv math.*, 9 Jun–20 Aug 2026 [‡]	3,207	3,207	–	15 MB	Post-release scrape + GSM8K shingle screen

Table 3: Corpora. All four receive identical exact-deduplication, Gopher quality filtering and MinHash/LSH near-duplicate removal at Jaccard 0.60; they differ in *contamination* control, deliberately. The reference-free screen ran on the news corpus, whose 8 June 2026 snapshot predates part of the cohort’s release window; Reddit and Hacker News were scraped after every model shipped, as was most of the arXiv set, and they are protected by construction. Reddit and Hacker News comments are grouped to document length. [†]Four days are available in the stated range. [‡]Used only for the mathematics adaptation.

B The Full Benchmark Alignment

Table 4 reports all twenty-four (corpus $\times$ reading $\times$ benchmark) cells over the full 55 pairs; Table 5 prices each reading’s top-1 pick against each benchmark’s own. The rotation (Appendix C) is legible column by column: adapting on general text raises HellaSwag agreement and lowers the GSM8K column on every corpus and MMLU-Pro on two of the three general-text corpora; adapting on mathematics reverses the GSM8K half. The pair columns show how little of this a rank correlation conveys: across its eight rows the GSM8K column spans 41 to 49 of 55 pairs.

C Robustness and Measurement Artefacts

Stability across domains. All eleven models ran on three independently collected domains under the same recipe (Tables 3 and 9). Adaptation genuinely changes the choice: unadapted and adapted rankings agree only at Spearman $\rho = 0.655$ on news, 0.609 on Reddit and 0.682 on Hacker News, with models moving up to seven places, and the adapted ordering then holds across domains: news and Reddit agree on nine of eleven exact ranks ($\rho = 0.991$, maximum displacement one), news against Hacker News gives 0.955 and Reddit against Hacker News 0.964, and the agreement is not merely ordinal (Figure 5). **So does the zero-shot ordering:** 0.982, 0.964 and 0.973 on the same three comparisons. The tempting story is that adaptation buys stability, and it does not; it buys a *different* stable ordering, and Section 3.1 is the evidence that it is the better of the two.

Corpus	Reading	Rank correlation (ρ) with			Pairs ordered the same way, of 55		
		GSM8K	MMLU-Pro	HellaSwag	GSM8K	MMLU-Pro	HellaSwag
News	zero-shot	0.800	0.845	0.718	45	45	44
	adapted	0.700	0.773	0.982	42	44	53
Reddit	zero-shot	0.745	0.782	0.655	43	43	42
	adapted	0.655	0.745	0.964	41	43	52
Hacker News	zero-shot	0.800	0.809	0.627	44	44	43
	adapted	0.745	0.855	0.973	44	46	53
Math (arXiv)	zero-shot	0.764	0.736	0.482	43	43	40
	adapted	0.909	0.927	0.818	49	49	46

Table 4: Adaptation turns the ranking toward the benchmark that matches the text. Rank correlation of BPB with each benchmark over the eleven models, before (zero-shot) and after adaptation, on every corpus; ρ is signed so that 1 means lower BPB goes with a higher score. On the three general corpora adaptation lifts HellaSwag from 0.63–0.72 to 0.96–0.98 while GSM8K falls; on arXiv mathematics it lifts GSM8K and MMLU-Pro instead. The last three columns count, of all 55 pairs of the eleven (not only size-matched ones), those that BPB orders the same way as each benchmark: adapted news BPB agrees with HellaSwag on 53.

Why the search gives every model the same number of trials. The search of Phase 2 uses a multivariate TPE sampler (seed 42) over learning rate (10^{-6} – 10^{-3} , log), physical batch (2, 4, 8), gradient accumulation (1–16), warm-up ratio (0–0.2), weight decay (10^{-5} – 10^{-1} , log), LoRA rank (8–64), α as a multiple of rank (1, 2, 4) and dropout (0–0.2); a successive-halving pruner starts trials at 50 steps and promotes the best third at each rung; and the objective is BPB on a fixed 256-block subset of the validation split. An earlier run of this search on the eleven-model news cohort was capped by wall-clock time rather than by trial count, and the cap was confounded with model size: the two largest models completed four trials each while the 1B models completed 17 and 18 (results/legacy_sweep/*/trials.csv), and the five models of 9B or more all kept the same third draw of the seeded sampler as their winner, while the smaller models’ winners came from much later in their searches. A time-capped search hands small models a better-tuned adapter and biases the ranking by parameter count, which is why the released pipeline budgets the search in trials and applies no time cap by default. Apart from the right panel of Figure 3, which uses three of its adapters to show where in a block adaptation gains, none of those adapters produces a number in the paper; the reported results use the hand-picked configuration of Phase 2.

Membership screens. The news corpus additionally passes three reference-free membership screens per model: Min-K++ (normalised likelihood of the lowest-probability $K=20\%$ target tokens), context sensitivity (target NLL with and without sampled same-corpus prefixes), and, for flagged candidates only, preference over two word-swap/placeholders perturbations. The global union quarantined 80 unique sequences, after which every model receives the identical retained corpus. Near-100% preference for the unperturbed original is expected (fluency alone produces it), so we draw no memorisation conclusion from it (results/contamination/).

The cold start, seen a second way. Varying the inference context on 64 fixed held-out target windows from 16 to 2048 tokens reproduces Figure 3’s effect non-positionally: both models improve with context and the larger improves faster, so any protocol that scores short contexts under-rates the models that use long ones. Measured context sensitivity is 0.032 bits per byte per doubling, which is the constant the block-alignment bound below uses.

Token-aligned blocks: a real bias, bounded. Evaluation blocks are 512 *tokens* rather than a fixed byte span, so models with different tokenizers score slightly different amounts of text: 5.92–6.24 MB across the eleven-model cohort, a 5% spread, and 9.9% across the seventeen. Bounding the resulting bias with the measured context sensitivity puts it at $0.08\times$ the smallest adjacent rank gap in the eleven-model cohort. In the extended cohort the bound is looser and worth stating: on the worst adjacent pair, Qwen2.5-3B over Falcon3-7B, whose BPB gap is only 0.29%, the bound reaches $0.61\times$ the gap. It still **overturns no rank**, but byte-aligned blocking would remove the question, and we recommend it for future work.

Corpus and reading	Picks	Accuracy given up vs. the benchmark’s own pick (pts)		
		GSM8K	MMLU-Pro	HellaSwag
News zero-shot	Qwen-3.5-35B-MoE	0.0	0.0	2.4
News adapted	Gemma-4-31B	1.6	4.3	0.0
Reddit zero-shot	Ministral-3-14B	5.8	8.6	5.0
Reddit adapted	Gemma-4-31B	1.6	4.3	0.0
Hacker News zero-shot	Qwen-3.5-35B-MoE	0.0	0.0	2.4
Hacker News adapted	Gemma-4-31B	1.6	4.3	0.0
Math (arXiv) zero-shot	Qwen-3.5-35B-MoE	0.0	0.0	2.4
Math (arXiv) adapted	Qwen-3.5-35B-MoE	0.0	0.0	2.4

Table 5: What choosing on BPB costs you on each benchmark. For every corpus and reading, the model the ranking puts first and how many accuracy points that costs against each benchmark’s own top-1 model. Only three distinct picks occur in eight rows, which is itself the finding: the adapted ranking selects Gemma-4-31B on every general-text corpus (the HellaSwag leader outright, at 1.6 GSM8K and 4.3 MMLU-Pro points), while adapting on mathematics recovers the GSM8K and MMLU-Pro leader at zero regret. The corpus, not the metric, decides which column you pay in.

Corpora compared	Rank agreement (Spearman)		Value agreement (Pearson)	
	unadapted	adapted	unadapted	adapted
News–Reddit	0.982	0.991	0.996	0.992
News–Hacker News	0.964	0.955	0.994	0.977
News–arXiv maths	0.909	0.755	0.964	0.839
Reddit–Hacker News	0.973	0.964	0.998	0.990
Reddit–arXiv maths	0.918	0.736	0.971	0.882
Hacker News–arXiv maths	0.973	0.836	0.982	0.930

Table 6: The ordering is a property of the models, not of one corpus. Agreement between the eleven models’ BPB on two corpora. The three general-text corpora rank the models almost identically before and after adaptation. Mathematics is the telling row: adaptation makes it agree *less* with the general corpora, because adapted on mathematics the ranking follows a different capability (Section 3.2).

Seed variation is negligible. Every adaptation cell ran with the torch RNG unseeded, so a preregistered check re-ran all 17 news models three times (51 cells, everything else pinned). Median within-model SD of adapted BPB is 0.000087 against a between-model SD of 0.074340, a ratio of 0.0012, and *no* pair of models within $2\times$ in size (0 of 40) reverses its ordering across seeds. The spread is not uniform: Gemma-4-12B is the one model where seed noise is visible at all (SD 0.00315, $5.3\times$ the next-largest spread).

Steering: direction, not magnitude. Adapting on general text raises agreement with HellaSwag on news from $\rho = 0.718$ to 0.982 while GSM8K falls from 0.800 to 0.700 and MMLU-Pro from 0.845 to 0.773; adapting on contamination-screened arXiv mathematics instead raises GSM8K agreement to 0.909 and restores the GSM8K oracle pick at zero regret, while three general-text corpora tried as controls all leave GSM8K at or below the do-nothing baseline. That is the rotation, and it is why the body reports only its direction. Because the steering hypothesis predicts a *joint* movement, the better-powered statistic is a difference in differences: (GSM8K minus HellaSwag selection accuracy under mathematics adaptation) minus the same quantity under a general-text corpus, which is $+0.242$ pooled. Resampling models gives a strictly positive difference in 93.7% of draws; the stricter publisher-family clustering the mechanism itself implies gives 80.9% with not one negative draw in 20,000, but no longer clears 0.05 on any tie convention. A leave-one-family-out jackknife shows why: dropping the two Gemma models takes the pooled difference from $+0.242$ to $+0.056$. The remedy is more *families*, not more models.

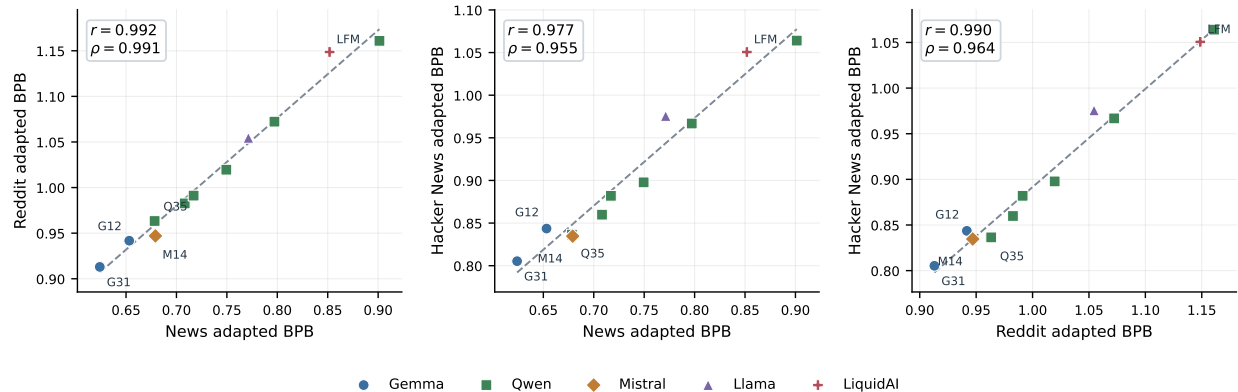


Figure 5: Adapted BPB across domains. Each point is one model, its adapted BPB on two corpora. Absolute difficulty changes, but points remain close to a monotone shared ordering: Pearson correlations between the values are 0.992 (news–Reddit), 0.977 (news–Hacker News) and 0.990 (Reddit–Hacker News). G31/G12 denote Gemma-4-31B/12B; Q35 is Qwen-3.5-35B-MoE; M14 is Ministral-3-14B.

Model	Out of the box			After	Adapted BPB
	raw loss	GSM8K	MMLU-Pro	fine-tuning	predicted
Gemma-4-12B	13	9	8	1	2
Llama-3.2-1B	10	15	15	11	11
LFM2.5-1.2B	15	12	13	14	14

Table 7: How a model looks is not how it finishes. Rank among the fifteen fine-tuned models (1 is best). The benchmarks misjudge where these models end up, and raw loss misjudges Gemma-4-12B by twelve places: it is mid-table on both benchmarks and near the bottom on raw loss, then builds the best system of all. Llama-3.2-1B is last on both benchmarks and finishes eleventh. Adapted BPB, measured before any task fine-tuning, places every model within a single rank of where it finishes: all fifteen, and eleven exactly.

Three reproduction controls. Re-running two Ministral-3-14B cells weeks later on different hardware reproduced zero-shot BPB exactly and adapted BPB to within 0.0003, two orders of magnitude below the differences compared. Because the extension models were adapted on a different software stack, a published cell (Qwen2.5-7B) was re-run through the extension pipeline; the analysis refuses to pool the cohorts unless it reproduces within 1%, and it reproduces to 0.01% on adapted BPB (0.7169 against 0.7170). And because benchmark rankings are themselves estimates, recomputing selection accuracy while discarding pairs whose benchmark gap is smaller than k combined standard errors leaves the result intact and slightly improves it: on GSM8K 0.909 at $k=0$ becomes 0.933 at $k=3$, and on HellaSwag 0.836 becomes 0.863, so the metric’s errors concentrate in pairs the benchmark itself cannot separate.

D The Downstream Fine-Tune

Section 3.4 reports the result; this is how it was built, and what it does and does not license. It ran in two stages, each preregistered in code before its job returned: first the two pairs on which the selectors disagree most (Gemma-4-12B against Ministral-3-14B, Llama-3.2-1B against Qwen-2.5-1.5B), then every cohort model that fits one GPU, so that no pair is chosen by anyone. Gemma-4-31B and Qwen-3.5-35B-MoE need more than one GPU *for this fine-tune* (its 1,280-token sequences and its generation pass cost far more memory than the 512-token adaptation priced in Section 1) and are excluded by that rule, which costs one in-band pair of forty.

The task. Journalistic convention makes a news article’s first paragraph a standalone summary of it, so the corpus supplies its own supervision: the input is the article with its opening paragraph removed, truncated to 3,000 characters, and the target is that paragraph. Articles qualify if they have at least three paragraphs, an opening paragraph of 200–

700 characters ending in sentence punctuation, and at least 800 characters of remaining body. Of the 11,905 validation documents, 3,627 qualify; 2,000 are sampled for training and 500 for evaluation under a fixed seed.

Why the validation split. The BPB test blocks live in the *test* split, so the fine-tuning task is built from the validation split instead. That is verified rather than assumed: the test split reproduces from an independent reconstruction, and the adaptation pipeline’s contamination pass (which would have split the corpus differently) is off by default and recorded as off in every published cell. Evaluation documents are then hashed against the test documents twice, on full normalised text and on a 400-character opening, the second because a syndicated wire story is republished with an edited body and an identical lede. Four articles were dropped by that second check and none by the first.

Equal preparation. Both members of a pair get the paper’s own adaptation recipe unchanged (LoRA rank 16, $\alpha=32$, dropout 0.05, learning rate 10^{-4} , effective batch 32, 250 steps) with prompt tokens masked out of the loss so the model is trained to write the lede rather than to reconstruct the article. Nothing is tuned on either model, because a per-model search is exactly the cost this protocol exists to avoid. The tokenizers differ, so the budget is matched in update steps $\times$ batch $\times$ sequence length; each model’s realised token count is in the artifact. Decoding is greedy with a 180-token cap and each checkpoint’s own generation configuration otherwise untouched, following Appendix F. The prompt is assembled so that the instruction always survives truncation: the *body* is cut to fit, never the instruction after it, because right-truncating the assembled prompt would delete the instruction from the longest articles at a different article count for each tokenizer: an asymmetry between the two models being compared. At most four of the 500 evaluation articles were cut for any model, and none for three of them.

The decision rules, fixed before each run. For the two chosen pairs: mean ROUGE-L F1 over the 500 held-out articles, a paired bootstrap over articles (20,000 draws, 95% percentile interval) and a practical-equivalence margin of 0.01; an interval excluding zero resolves the pair for whichever reading predicted that winner. The first pair resolved for adapted BPB; the second was inconclusive. For the full cohort: the fine-tuned ROUGE-L is the outcome and every selector is scored with the paper’s own statistic: pairwise accuracy within $2\times$, model-level cluster bootstrap, 20,000 draws (Table 2). A separate guard voids any comparison in which fine-tuning failed to improve a model over its own base weights, since ordering unmoved models would order their base behaviour under another name. It did not fire: all fifteen models gained between $+0.041$ and $+0.153$.

A second recipe. Both readers of an earlier draft raised the same objection: the fine-tune reused the selector’s own recipe, so a model that adapts quickly under it could win both. Recipe B changes every setting the two shared (LoRA rank 64, $\alpha=128$, learning rate 3×10^{-4} , effective batch 16, 400 steps, a new seed) on the same task, data, decoding and metric, preregistered before it ran. It fits the training articles far harder and scores slightly below recipe A for every model, yet orders the fifteen systems almost identically (Spearman 0.982). Against recipe-B systems adapted BPB remains the top selector at 0.923 [0.750, 1.000], ahead of HellaSwag at 0.846, parameter count at 0.744 and GSM8K and MMLU-Pro near chance, and it picks the better system on all 20 pairs recipe B resolves. Its margin over parameter count, resolved under recipe A, does not resolve here ($+0.179$ [$-0.036, +0.441$]), so that margin should be read as recipe-specific (Table 2).

What copying buys. A news body restates its own lede, so reference overlap is partly available by retrieval. Two reference points bound it: the article’s first body sentence scores 0.125 against the reference and a random body sentence 0.111. Unadapted base weights land close to copying (inspection confirms they largely do copy), while every fine-tuned system clears it, from 0.188 (Qwen-2.5-0.5B) to 0.249 (Gemma-4-12B).

How close the corpus has to be. Exploratory, run after the result: scoring the news fine-tune with adapted BPB from each corpus, on the ten size-matched pairs of the nine fine-tuned models that have all four, gives 0.900 for news and for Reddit, 0.700 for Hacker News and 0.600 for arXiv mathematics; the unadapted readings reach 0.500–0.600. Fresh general text carries the prediction, and mathematics, which aims the ranking at reasoning (Section 3.2), carries it least. Ten pairs support no interval, so this is a direction, not a result.

What this does not show. One task, one domain, two recipes, one seed each. On this task adapted BPB ordered the fine-tuned systems better than any other selector and GSM8K and MMLU-Pro ordered them at chance; its margin over HellaSwag does not resolve. It is not a claim that adapted BPB predicts downstream utility in general, and the paper does not make one.

E The Cohorts

Table 8 is the extended 17-model cohort behind the seed replicates and, less the two models too large for one GPU, the fine-tune: every model, its parameter count and its adapted news BPB. Table 9 then gives the original eleven across all three general corpora and both BPB endpoints, which is the population every other analysis in the paper uses. Table 9

Model	Params (B)	Adapted BPB	Cohort
Gemma-4-31B	31.4	0.624	original
Mistral-Nemo-12B	12.3	0.646	added
Gemma-4-12B	12.0	0.653	original
Qwen-2.5-14B	14.8	0.675	added
Qwen-3.5-35B-MoE*	34.7	0.678	original
Ministral-3-14B	14.0	0.679	original
Qwen-3.5-9B	9.0	0.708	original
Qwen-2.5-7B	7.7	0.717	original
Falcon3-10B	10.4	0.728	added
Qwen-3.5-4B	4.2	0.749	original
Qwen-2.5-3B	3.1	0.756	added
Falcon3-7B	7.5	0.758	added
Llama-3.2-1B	1.2	0.771	original
SmolLM2-1.7B	1.7	0.788	added
Qwen-2.5-1.5B	1.6	0.797	original
LFM2.5-1.2B	1.2	0.852	original
Qwen-2.5-0.5B	0.5	0.901	original

Table 8: The extended cohort, in full. All 17 models ordered by adapted news BPB. “added” marks the six models admitted for the size-matched extension on preregistered size-adjacency and publisher-spread criteria; the other eleven carry the four-corpus analyses. *Qwen-3.5-35B-MoE activates 3B of its 34.7B parameters; parameter count here is always the total.

is the full three-corpus leaderboard: both endpoints for all eleven models on news, Reddit and Hacker News, ordered by adapted news BPB. It is the artifact every eleven-model number in the paper derives from, as Table 8 is for the seventeen.

Three checkpoints are excluded. The two OLMo-2 models fell under the preregistered infrastructure-fault rule: their adapted BPB cells exist and are unused, because the generation engine the study’s evaluations share would not initialise for them, identically before and after a cache rebuild. The third, Qwen-2.5-32B, did not finish its 250 steps inside the run window; it would have added about two size-matched pairs at the top of the size range. The OLMo exclusion removed a whole publisher, which is what the cohort most needs.

F The Benchmark Column Re-Score

Harness configuration. MMLU-Pro is 5-shot chain-of-thought with exact match on one stratified 1,000-question subset (seed 42); HellaSwag is 10-shot length-normalised multiple-choice accuracy on all 10,042 validation examples; GSM8K is 5-shot chain-of-thought with flexible numeric extraction on all 1,319 test examples. All bf16, greedy, tokenizer-native BOS.

July’s eleven-model column was scored on a host that no longer exists and generative scores do not pool across hosts: identical released weights move -0.60 to $+0.69$ GSM8K points when only the host changes, A100 to H100,

Model	News		Reddit		Hacker News	
	zero-shot	adapted	zero-shot	adapted	zero-shot	adapted
Gemma-4-31B	0.752	0.624	1.062	0.913	0.962	0.805
Gemma-4-12B	0.866	0.653	1.172	0.942	1.063	0.844
Qwen-3.5-35B-MoE	0.707	0.678	1.008	0.963	0.866	0.836
Minstral-3-14B	0.718	0.679	1.008	0.947	0.876	0.835
Qwen-3.5-9B	0.740	0.708	1.039	0.983	0.894	0.860
Qwen-2.5-7B	0.758	0.717	1.051	0.991	0.922	0.882
Qwen-3.5-4B	0.782	0.749	1.080	1.020	0.933	0.898
Llama-3.2-1B	0.811	0.771	1.109	1.055	1.011	0.976
Qwen-2.5-1.5B	0.833	0.797	1.128	1.072	1.002	0.967
LFM2.5-1.2B	1.340	0.852	1.779	1.149	1.687	1.051
Qwen-2.5-0.5B	0.935	0.901	1.222	1.161	1.099	1.064

Table 9: Marker-masked BPB under one fixed adaptation protocol. Lower is better. Rows are ordered by adapted news BPB; all eleven models occur in all three corpora.

so it was re-scored in full: seventeen models $\times$ three benchmarks = fifty-one cells, one host (A100-SXM-64GB), one build, with the design and the exclusion rule fixed in advance. Task definitions match exactly (17/17 identical task hashes on both hashed benchmarks) and every overlapping cell agrees with July to within 1.3 points, most to within 0.3.

Reaching that took one correction worth reporting, because the fault was in our own fix. Because the newer backend honours each checkpoint’s shipped `generation_config.json` and the original did not, we replaced every checkpoint’s generation settings with one identical policy, which also discarded `suppress_tokens`, a setting that is live for gemma-4-12B. That single omission cost it 9.86 GSM8K and 14.30 MMLU-Pro points while leaving HellaSwag, scored by likelihood and never calling `generate()`, intact. Keeping the setting for all seventeen models reproduces July to +1.21 points on GSM8K, +0.10 on MMLU-Pro and -0.27 on HellaSwag. That GSM8K cell is the one outside the preregistered 1.0-point tolerance, within the allowance of one that the re-score declared in advance: the column is within its stated allowance rather than clean, and those are different claims. The lesson generalises: a wrapper imposed to remove a confound can be the confound, so the unmodified harness belongs in the design as a control.